

M – Tech Production & Marketing

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REPORT OF EXPERIMENT 10

Computer Vision — CNNs & OpenCV

Introduction

This report summarizes the activities completed during **Experiment 10** of the M-Tech Production & Marketing AI/ML Internship Program. The experiment focused on understanding the fundamentals of **Computer Vision** and performing basic image processing operations using the OpenCV library. Practical activities included reading an image, converting it to grayscale, resizing, blurring, edge detection, and saving the processed images. These concepts provide a foundation for developing computer vision applications.

Objectives

- To understand the basics of Computer Vision.
- To learn image processing using OpenCV.
- To read and display image information.
- To perform grayscale conversion.
- To resize and blur images.
- To perform edge detection and to save processed images.

Task Description

The assigned task was to develop a Python application using OpenCV to perform basic image processing operations. The application loads an image, displays its dimensions,

converts it to grayscale, resizes it, applies Gaussian Blur, detects edges using the Canny algorithm, and saves all processed images for further analysis.

Work Done

Today, I developed a Computer Vision application using Python and OpenCV. I loaded an image, displayed its properties, and performed several image processing operations including grayscale conversion, image resizing, Gaussian Blur, and edge detection. After processing, I saved all generated images and verified that each operation produced the expected results. This practical activity helped me understand the fundamentals of image processing used in AI and Computer Vision applications.

Challenges Faced

During the experiment, I encountered minor issues related to selecting the correct image path and installing the OpenCV library. These issues were resolved by placing the image in the project folder, verifying the file name, and installing the required package correctly. After testing, all image processing operations worked successfully.

Key Learnings

This experiment improved my understanding of Computer Vision and OpenCV. I learned how to load images, extract image properties, convert images to grayscale, resize images, apply Gaussian Blur, detect edges, and save processed images. These techniques are widely used in image recognition, object detection, and other AI-based computer vision applications.

Conclusion

The experiment was completed successfully, and all objectives were achieved. The practical implementation strengthened my understanding of image processing using OpenCV and provided hands-on experience with basic Computer Vision techniques. This experiment serves as an important foundation for developing more advanced AI-based vision systems.

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